SUMMATIVE ASSESSMENT – 1 (SA-1) MODEL PAPER – SET 2

CLASS X · MATHEMATICS · AP State Syllabus 2026-27 · SSC Blueprint 15E & 16E Pattern

BLUEPRINT – SA-1 MODEL PAPER – SET 2

SUBJECT: MATHEMATICS	CLASS: X	TIME: 3 Hrs 15 Min	MAX. MARKS: 100

Syllabus: Chapters 1 – 11 of Class X Mathematics (AP SCERT, 2026-27) – Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations, Arithmetic Progressions, Triangles, Coordinate Geometry, Statistics, Introduction to Trigonometry, Applications of Trigonometry, Probability. Pattern: SSC Public Examinations 2026-27 Blueprint & Model Paper (15E & 16E).

1. Weightage to Academic Standards (Ability)

ABILITY	AWARENESS (60%)	SENSITIVITY (20%)	CREATIVITY (20%)	TOTAL
MARKS	60	20	20	100

2. Weightage to Form of Questions

FORM OF QUESTION	Essay / Long Ans.	Short Answer	Very Short Ans.	Objective	TOTAL
No. of questions	5	8	8	12	33
Marks allotted	8	4	2	1	100
Estimated time	70 min	60 min	30 min	20 min	180 + 15 min

3. Ability-wise Distribution of Questions

ABILITY	1 Mark	2 Marks	4 Marks	8 Marks	Total Marks
Awareness	4	4	6	3	60
Sensitivity	4	2	1	1	20
Creativity	4	2	1	1	20
Total Questions	12	8	8	5	100

4. Chapter-wise Weightage

S.No	UNIT / CHAPTER	MARKS
1	Real Numbers	1+8 = 9
2	Polynomials	1+2+4 = 7
3	Pair of Linear Equations	1+2+8 = 11
4	Quadratic Equations	1+2+4+(8) = 7 + (8)
5	Arithmetic Progressions	1+4+(8) = 5 + (8)
6	Triangles	1+1+2+4+(8) = 8 + (8)
7	Introduction to Trigonometry	1+2+4+(8) = 7 + (8)
8	Applications of Trigonometry	1+2+(8) = 3 + (8)
9	Coordinate Geometry	1+2+4+8 = 15
10	Statistics	1+2+4+8 = 15
11	Probability	1+4+8 = 13
	TOTAL	**100 + (40)**

Note: Marks within ( ) are the internal-choice (OR) questions of Section IV. The marks of the chapters that are not part of SA-1 in the SSC blueprint (Circles, Areas Related to Circles, Surface Areas & Volumes) are redistributed among Chapters 1 – 11; the ability-wise (60 : 20 : 20) and form-wise weightage are exactly as in the SSC blueprint.

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5. Question-wise Chapter Distribution and Domain (Questions 1 to 33)

Q.No	Chapter / Topic	Domain	Marks
1	Real Numbers	Awareness	1
2	Polynomials	Creativity	1
3	Pair of Linear Equations	Creativity	1
4	Quadratic Equations	Creativity	1
5	Arithmetic Progressions	Sensitivity	1
6	Triangles	Awareness	1
7	Introduction to Trigonometry	Sensitivity	1
8	Applications of Trigonometry	Sensitivity	1
9	Coordinate Geometry	Creativity	1
10	Statistics	Awareness	1
11	Probability	Awareness	1
12	Triangles	Sensitivity	1
13	Polynomials	Creativity	2
14	Quadratic Equations	Sensitivity	2
15	Triangles	Awareness	2
16	Coordinate Geometry	Sensitivity	2
17	Introduction to Trigonometry	Awareness	2
18	Statistics	Awareness	2
19	Pair of Linear Equations	Awareness	2
20	Applications of Trigonometry	Creativity	2
21	Polynomials	Awareness	4
22	Quadratic Equations	Awareness	4
23	Arithmetic Progressions	Awareness	4
24	Introduction to Trigonometry	Awareness	4
25	Triangles	Sensitivity	4
26	Coordinate Geometry	Awareness	4
27	Statistics	Awareness	4
28	Probability	Creativity	4
29a / 29b	(a) Real Numbers (b) Arithmetic Progressions	Creativity	8
30a / 30b	(a) Statistics (b) Triangles	Awareness	8
31a / 31b	(a) Coordinate Geometry (b) Introduction to Trigonometry	Sensitivity	8
32a / 32b	(a) Probability (b) Quadratic Equations	Awareness	8
33a / 33b	(a) Pair of Linear Equations (b) Applications of Trigonometry	Awareness	8

ABILITIES AND THEIR OBJECTIVES AS PER PARAKH NORMS

I) AWARENESS: Knowledge / understanding of the construct / understanding of the process (resembles old objectives Knowledge, Understanding and Application).

II) SENSITIVITY: Conflict resolution / open-mindedness / collaboration (resembles old objective Analysis).

III) CREATIVITY: Generate innovative ideas / flexibility / fluency / exploration / combining ideas and concepts (resembles old objectives Evaluation and Creativity).

NOTE: Overall, the chapter-wise and ability-wise weightage is followed. There is no rigid rule that a question with a particular number must come from a particular chapter.

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SUMMATIVE ASSESSMENT – 1 : MODEL QUESTION PAPER – SET 2

MATHEMATICS – CLASS X (English Medium, AP State Syllabus 2026-27)

Time : 3 Hrs 15 Min (15 min reading + 3 hrs writing) **Max. Marks : 100**

General Instructions: *(i) Answer all the questions. (ii) Write the answers neatly in the answer booklet. (iii) The paper has four Sections (I, II, III, IV). (iv) Use of calculators is not permitted. (v) Draw neat, labelled figures wherever necessary.*

SECTION – I 12 × 1 = 12 Marks

Note : *i) Answer all the questions. ii) Each question carries 1 mark. iii) Choose the correct option and write its letter (A/B/C/D) in the answer booklet.*

1. The HCF of 96 and 404 is *[Awareness · Real Numbers]*

(A) 2 (B) 4 (C) 8 (D) 12

2. If one zero of the polynomial $(a^2 + 9)x^2 + 13x + 6a$ is the reciprocal of the other, then a = *[Creativity · Polynomials]*

(A) 0 (B) 3 (C) −3 (D) 6

3. If the lines $3x + 2ky - 2 = 0$ and $2x + 5y + 1 = 0$ are parallel, then k = *[Creativity · Pair of Linear Equations]*

(A) 5/4 (B) 2/5 (C) 15/4 (D) 3/2

4. The roots of the equation $x^2 - 3x - m(m + 3) = 0$, where m is a constant, are *[Creativity · Quadratic Equations]*

(A) m, m + 3 (B) −m, m + 3 (C) m, −(m + 3) (D) −m, −(m + 3)

5. Priya jogs 200 m on the first day and increases the distance by 50 m every day. The distance she jogs on the 10th day is *[Sensitivity · Arithmetic Progressions]*

(A) 600 m (B) 650 m (C) 700 m (D) 750 m

6. If $\Delta ABC \sim \Delta DEF$, AB = 6 cm and DE = 9 cm, then BC : EF = *[Awareness · Triangles]*

(A) 3 : 2 (B) 2 : 3 (C) 4 : 9 (D) 9 : 4

7. If $3 \tan\theta = 4$, then $\sin\theta + \cos\theta$ = *[Sensitivity · Introduction to Trigonometry]*

(A) 7/5 (B) 1/5 (C) 12/5 (D) 5/7

8. The length of the shadow of a 20 m tall pole is $20\sqrt{3}$ m. The angle of elevation of the sun is *[Sensitivity · Applications of Trigonometry]*

(A) 30° (B) 45° (C) 60° (D) 90°

9. If A(1, 2), B(4, y), C(x, 6) and D(3, 5) are the vertices of a parallelogram ABCD taken in order, then *[Creativity · Coordinate Geometry]*

(A) x = 6, y = 3 (B) x = 3, y = 6 (C) x = 2, y = 4 (D) x = 4, y = 2

10. The class mark of the class interval 15–25 is *[Awareness · Statistics]*

(A) 10 (B) 20 (C) 40 (D) 5

11. Two coins are tossed simultaneously. The probability of getting at least one head is *[Awareness · Probability]*

(A) 1/4 (B) 1/2 (C) 3/4 (D) 1

12. In ΔABC, DE ∥ BC with AD = 4x − 3, DB = 3x − 1, AE = 8x − 7 and EC = 5x − 3. Then x = *[Sensitivity · Triangles]*

(A) 1 (B) 2 (C) 3 (D) 4

SECTION – II 8 × 2 = 16 Marks

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Note : *i) Answer all the questions. ii) Each question carries 2 marks.*

13. Find a quadratic polynomial whose zeroes are $2 + \sqrt{3}$ and $2 - \sqrt{3}$. *[Creativity · Polynomials]*

14. Ramu says that the equation $2x^2 - 6x + 9/2 = 0$ has two distinct real roots, but Somu says it has equal roots. Who is correct? Find the roots. *[Sensitivity · Quadratic Equations]*

15. State the SAS criterion for similarity of two triangles. *[Awareness · Triangles]*

16. Find the value(s) of x if the distance between the points (x, 7) and (1, 15) is 10 units. *[Sensitivity · Coordinate Geometry]*

17. Evaluate: $\tan 45°/\operatorname{cosec} 30° + \sec 60°/\cot 45° - (5 \sin 90°)/(2 \cos 0°)$. *[Awareness · Introduction to Trigonometry]*

18. Write the step-deviation formula for the mean of grouped data and explain the meaning of each symbol used. *[Awareness · Statistics]*

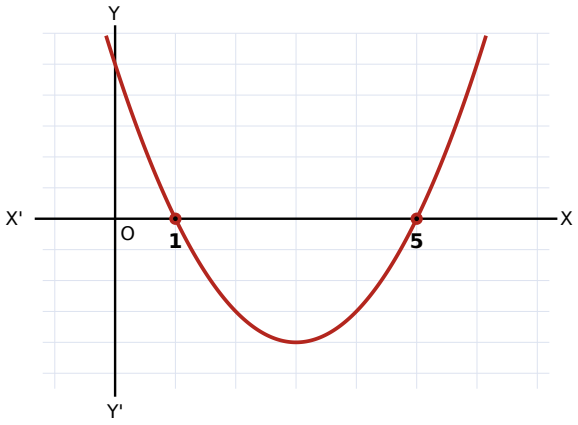
19. Solve by the elimination method: $5x - 3y = 11$ and $3x + 2y = 18$. *[Awareness · Pair of Linear Equations]*

20. The angle of elevation of the top of a 15 m tall pole from a point on the ground is 30°. Draw a rough figure and find the distance of the point from the foot of the pole. *[Creativity · Applications of Trigonometry]*

SECTION – III 8 × 4 = 32 Marks

Note : *i) Answer all the questions. ii) Each question carries 4 marks.*

21. Observe the graph of a polynomial given below and answer the questions. *[Awareness · Polynomials]*



i) How many zeroes does the polynomial have?
ii) Write the zeroes of the polynomial.
iii) Find the sum and the product of the zeroes.
iv) If the polynomial is $x^2 + bx + c$, find the values of b and c.

22. Find two numbers whose sum is 27 and product is 182. *[Awareness · Quadratic Equations]*

23. Which term of the AP 2, 7, 12, ... is 147? Also find the sum of its first 20 terms. *[Awareness · Arithmetic Progressions]*

24. Prove that: $(\operatorname{cosec}\theta - \sin\theta)(\sec\theta - \cos\theta)(\tan\theta + \cot\theta) = 1$. *[Awareness · Introduction to Trigonometry]*

25. A girl of height 90 cm walks away from the base of a lamp-post at a speed of 1.2 m/s. The lamp is 3.6 m above the ground. Find the length of her shadow after 4 seconds. *[Sensitivity · Triangles]*

26. Find the ratio in which the y-axis divides the line segment joining the points (5, −6) and (−1, −4). Also find the point of division. *[Awareness · Coordinate Geometry]*

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27. The marks obtained by 50 students in a test are given below. Find the mode of the data. *[Awareness · Statistics]*

Marks	0–10	10–20	20–30	30–40	40–50
No. of students	5	8	15	12	10

28. A box contains 20 cards numbered 1 to 20. One card is drawn at random. Write any four different events based on this experiment and find the probability of each. *[Creativity · Probability]*

SECTION – IV 5 × 8 = 40 Marks

Note : *i) Answer all the questions. ii) Each question carries 8 marks. iii) There is an internal choice for each question.*

29.

a) Show that the square of any positive integer is of the form 3m or 3m + 1 for some integer m, but never of the form 3m + 2. *[Creativity · Real Numbers]*

OR

b) The 4th term of an AP is zero. Prove that its 25th term is three times its 11th term. *[Creativity · Arithmetic Progressions]*

30.

a) Find the mean of the following data by the step-deviation method. *[Awareness · Statistics]*

Marks	0–20	20–40	40–60	60–80	80–100
No. of students	6	8	12	9	5

OR

b) State and prove the Pythagoras theorem. *[Awareness · Triangles]*

31.

a) A farmer has a field ABCD with corner points A(−5, 7), B(−4, −5), C(−1, −6) and D(4, 5). He wants to know its area. Divide the field into two triangles by the diagonal AC and find the total area of the field. *[Sensitivity · Coordinate Geometry]*

OR

b) Prove that: $(\sin\theta + \operatorname{cosec}\theta)^2 + (\cos\theta + \sec\theta)^2 = 7 + \tan^2\theta + \cot^2\theta$. *[Sensitivity · Introduction to Trigonometry]*

32.

a) A spinning wheel has 12 equal sectors numbered 1 to 12. The arrow is spun and stops at a sector at random. Find the probability that the number it stops at is: *[Awareness · Probability]*

i) divisible by 3
ii) a composite number
iii) a prime number
iv) greater than 9

OR

b) Rohan's mother is 26 years older than him. The product of their ages 3 years from now will be 360. Find Rohan's present age. *[Awareness · Quadratic Equations]*

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33.

a) Draw the graph of the pair of linear equations $x + 2y = 8$ and $2x - y = 1$ and find the solution from the graph. *[Awareness · Pair of Linear Equations]*

OR

b) A man standing on the bank of a river observes that the angle of elevation of the top of a tree on the opposite bank, directly opposite to him, is 60°. When he moves 40 m away from the bank, the angle of elevation becomes 30°. Find the height of the tree and the width of the river. *[Awareness · Applications of Trigonometry]*

— ALL THE BEST —

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